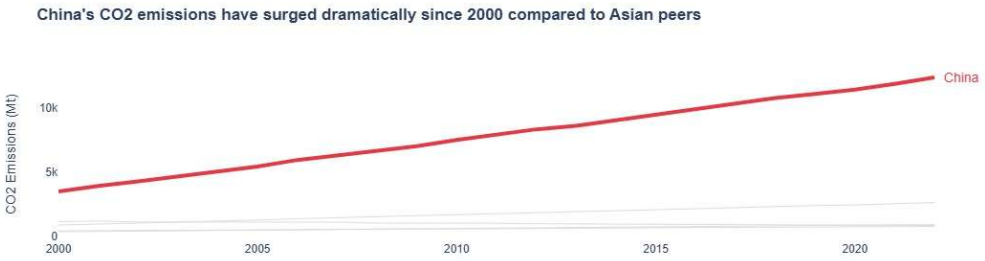
[6]:
```
import pandas as pd import plotly.express as px import
plotly.graph_objects as go df = pd.read_csv('co2_emissions (1).csv')
print(f"Loaded: {len(df)} rows | Countries:
{df['Country'].nunique()} | Years:␣
↪{df['Year'].min()}-{df['Year'].max()}")
print(df.head())
```

```
Loaded: 345 rows | Countries: 15 | Years: 2000-2022
       Country       Region Year CO2_Mt CO2_per_capita

0 United       North        2000 5857.6           1.32
  States       America
1 United       North        2001 5724.0           1.26
  States       America
2 United       North        2002 5652.8           1.11
  States       America
3 United       North        2003 5592.8           1.29
  States       America
4 United       North        2004 5743.2           1.12
  States       America
```

[7]:
```
  print("Countries:",   df['Country'].unique())   print("\nCO2
range:",  df['CO2_Mt'].min(),  "to",  df['CO2_Mt'].max(),  "Mt")
print("\nRegional                averages                (2022):")
print(df[df['Year']==2022].groupby('Region')['CO2_Mt'].mean().
↪sort_values(ascending=False).round(1))
```

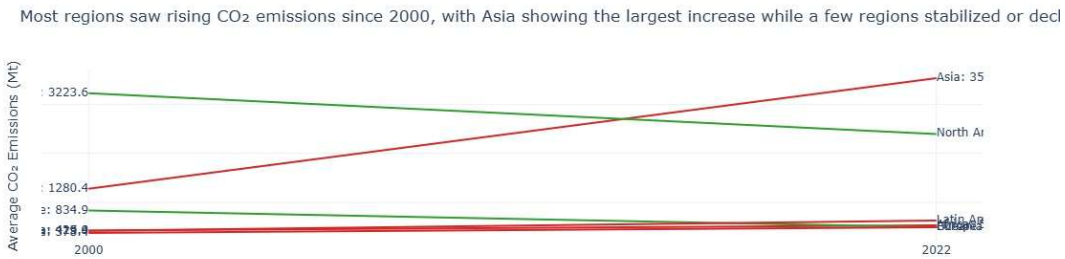
```
Countries: ['United States' 'China' 'India' 'Germany' 'United
Kingdom' 'France'
 'Brazil' 'Japan' 'Canada' 'Australia' 'South Korea' 'Russia'
 'South Africa' 'Mexico' 'Indonesia']

CO2 range: 125.3 to 12409.5 Mt

Regional averages (2022):
Region
Asia          3531.1
North America 2393.8
Latin America  629.2
Africa         534.4
Europe         496.5
Oceania        493.7
Name: CO2_Mt, dtype: float64
```

[9]:
```
import plotly.graph_objects as go
df_asia = df[df['Region'] == 'Asia']
countries = df_asia['Country'].unique()
target = 'China'
fig = go.Figure()
for country in countries:
    country_data = df_asia[df_asia['Country'] == country]

    if country == target:
        color = '#E63946'
        width = 4
        show_label = True
    else:
        color = '#DDDDDD'
        width = 1
        show_label = False

    fig.add_trace(go.Scatter(
        x=country_data['Year'],
        y=country_data['CO2_Mt'],
        mode='lines',
        name=country,
        line=dict(color=color, width=width),
        hoverinfo='name+y'
    ))
last_year = df_asia['Year'].max()
last_val = df_asia[(df_asia['Country'] == target) & (df_asia['Year'] ==␣
 ↪last_year)]['CO2_Mt'].values[0]

fig.add_annotation(
    x=last_year, y=last_val, text=f"  {target}",
    showarrow=False, xanchor="left", font=dict(color='#E63946', size=14,␣
 ↪family="Arial")
)

fig.update_layout(
    title="<b>China's CO2 emissions have surged dramatically since 2000␣
 ↪compared to Asian peers</b>",
    xaxis_title="", yaxis_title="CO2 Emissions (Mt)",
    plot_bgcolor='white', showlegend=False,
    font=dict(family="Arial", size=12),
    margin=dict(r=100)
)

fig.show()
```


China's CO2 emissions have surged dramatically since 2000 compared to Asian peers
China
CO2 Emissions (Mt)
10k
5k
0
2000
2005
2010
2015
2020

[10]:
```
import plotly.graph_objects as go
region_year = (df.groupby(['Region', 'Year'])['CO2_Mt']
               .mean()
               .reset_index())
filtered = region_year[region_year['Year'].isin([2000, 2022])]
pivot = filtered.pivot(index='Region', columns='Year', values='CO2_Mt').dropna()

fig = go.Figure()
for region in pivot.index:
    y0 = pivot.loc[region, 2000]
    y1 = pivot.loc[region, 2022]
    color = '#d62728' if y1 > y0 else '#2ca02c'
    fig.add_trace(go.Scatter(
        x=[2000, 2022],
        y=[y0, y1],
        mode='lines',
        line=dict(color=color, width=2),
        showlegend=False
    ))
    fig.add_trace(go.Scatter(
        x=[2000],
        y=[y0],
        mode='text',
        text=[f"{region}: {y0:.1f}"],
        textposition='middle left',
        showlegend=False
    ))
    fig.add_trace(go.Scatter(
        x=[2022],
        y=[y1],
        mode='text',
        text=[f"{region}: {y1:.1f}"],
        textposition='middle right',
        showlegend=False
```

```
    ))
fig.update_layout(
    template='plotly_white',
    title='Most regions saw rising CO emissions since 2000, with Asia showing
 ↪the largest increase while a few regions stabilized or declined ',
    xaxis=dict(
        tickvals=[2000, 2022],
        title=''
    ),
    yaxis=dict(
        showticklabels=False,
        title='Average CO Emissions (Mt)  '
    )
)

fig.show()
```



[ ]:

```

```